

SETS

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Empty sets – An interesting analogy for understanding the difference between $\emptyset$ and $\{\emptyset\}$ is to think of sets as **boxes**. Think of $\emptyset$ as an **empty box**—there is nothing inside it. Now think of $\{\emptyset\}$ as a *box containing one empty box*. The outer box is not empty because it contains the inner empty box.

Cartesian Product – The inequality $(A \times B) \times C \neq A \times B \times C$ comes down to a difference in the **structure of their elements** (the nesting of the parentheses):

- $(A \times B) \times C$ **contains ordered pairs**: Its elements look like $((a, b), c)$, where the first component is itself an ordered pair.
- $A \times B \times C$ **contains ordered triples**: Its elements look like (a, b, c) , which are flat 3-tuples.

Because a nested pair like $((a, b), c)$ is formally distinct from a 3-tuple like (a, b, c) , the two sets consist of entirely different mathematical objects and are strictly unequal. *While strictly unequal, they are naturally isomorphic or in a 1-to-1 correspondence, which is why they are often treated interchangeably in informal contexts.*

Power set – If A is a set, the **power set** of A is another set, denoted as $\mathcal{P}(A)$ and defined to be the set of all subsets of A . In symbols, $\mathcal{P} = \{X : X \subseteq A\}$. If A is a finite set, then $|\mathcal{P}(A)| = 2^{|A|}$.

Well-Ordering Principle – Any non-empty subset of $\mathbb{N}$ has a smallest element.

Russell's Paradox – The philosopher and mathematician Bertrand Russell (1872–1970) did groundbreaking work on the theory of sets and the foundations of mathematics. He was probably among the first to understand how the misuse of sets can lead to bizarre and paradoxical situations. He is famous for an idea that has come to be known as Russell's paradox.

Russell's paradox involves the following set of sets:

$$A = \{X : X \text{ is a set and } X \notin X\}$$

Most sets we can think of are in A . The set $\mathbb{Z}$ of integers is not an integer (i.e., $\mathbb{Z} \notin \mathbb{Z}$) and therefore $\mathbb{Z} \in A$. Also $\emptyset \in A$ because $\emptyset$ is a set and $\emptyset \notin \emptyset$.

Is there a set that is not in A ? Consider $B = \{\underbrace{\{\{\{\dots\}\}\}}_B\}$. Think of B as a box containing a box, containing a box, containing a box, and so on, forever.

$$B = \underbrace{\{\{\{\{\dots\}\}\}\}}_B$$

Thus $B \in B$. As B does not satisfy $B \notin B$, it doesn't belong to A .

Russell's paradox arises from the question "**Is A an element of A ?**". For set X , $X \in A$ means the same thing as $X \notin X$. So for $X = A$, the previous line says $A \in A$ means the same thing as $A \notin A$. This is the Russell's paradox.

The paradox instigated a very careful examination of set theory and an evaluation of what can and cannot be regarded as a set. Eventually mathematicians settled upon a collection of axioms for set theory—the so-called **Zermelo-Fraenkel axioms**.

The Core Axioms of Zermelo-Fraenkel axioms:

The theory is defined by foundational axioms that govern set construction and behavior:

- **Axiom of Extensionality:** Two sets are equal if and only if they contain the exact same elements.
- **Axiom of Regularity (Foundation):** Every non-empty set x contains an element disjoint from x , which rules out circular membership chains like $x \in x$.
- **Axiom Schema of Specification (Separation):** Subsets can be formed by filtering elements of an existing set using a logical property.
- **Axiom of Pairing:** For any two sets, there exists a set containing precisely those two elements.
- **Axiom of Union:** For any set of sets, there exists a union containing all elements from the collection.
- **Axiom of Power Set:** For any set, there exists a set containing all of its subsets.
- **Axiom of Infinity:** There exists an inductive set containing the empty set and all its successive elements.
- **Axiom Schema of Replacement:** Applying a definable function to the elements of a set yields another valid set.
- **Axiom of Choice:** For any collection of non-empty sets, a choice function exists that can simultaneously select an element from each set.

ZFC provides a universal language capable of formalizing nearly all contemporary mathematics. However, under Gödel's incompleteness theorems, a consistent system like ZFC cannot prove its own consistency, and certain propositions—such as the Continuum Hypothesis—are independent of the axioms.
